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Dear Editors of Neural Networks,

Please consider our manuscript, “Quantization fragility under image corruption is recipe-dependent: paired evidence from INT8 and FP8 object detectors,” for publication as a research article in Neural Networks.

Low-precision neural networks are increasingly deployed on edge hardware, and practitioners often ask whether INT8 or FP8 is more robust to corrupted inputs. We show that the answer depends on how the engine is built and which part of the network is quantized, not on the datatype label. Our protocol pairs every corrupted input with a JPEG-95 clean control, feeds identical bytes to both precisions, and estimates the corruption-induced change in the FP8-INT8 average-precision gap inside common-image bootstrap samples.

The paper makes three main points. First, across nine YOLO11 dataset-by-scale blocks on KITTI, VOC, and a 2,000-image COCO subset, this interaction changes sign, while FP8 still has higher absolute AP in every block. Second, a matched-recipe decomposition of RetinaNet traces a severe INT8 collapse (21.2 AP on KITTI) to the regression head, and an operand factorial attributes almost all of it to activation quantization. Keeping that head in floating point repairs the deficit without retraining. FCOS shows no such deficit, and both results recur on off-the-shelf COCO-pretrained checkpoints. The pattern also replicates on a KITTI holdout scored with a checkpoint retrained without those images. Third, a two-fold corruption-aware calibration experiment with codec-matched controls helps in one fold-by-dataset cell and significantly hurts another, so it is not a reliable repair.

The work fits the scope of Neural Networks because it studies how quantization interacts with the internal structure of trained networks: it locates the fragile network region and the operand responsible, rather than ranking formats.

The manuscript is original, is not under consideration elsewhere, and all six authors have approved its submission. The research received no specific grant from funding agencies in the public, commercial, or not-for-profit sectors. Code, manuscript sources, and hash-bound evidence are archived in release v3.0.4 (version DOI 10.5281/zenodo.23097634) under concept DOI 10.5281/zenodo.22031663. Raw datasets, checkpoints, built engines, ONNX graphs, and raw predictions are not redistributed; their provenance and regeneration steps are documented. The manuscript includes data and code availability, funding, competing-interest, CRediT, and generative-AI declarations.

Thank you for your consideration.

Sincerely,

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